

SOCIAL MEDIA REACH ANALYSIS

INTERNSHIP PROJECT REPORT

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Domain: Data Analytics / Social Media Optimization

1. INTRODUCTION

In today's digital-first landscape, understanding social media performance metrics is vital for content creators, brand strategists, and marketing teams. Reach represents the number of unique users who view content across social platforms. Analyzing reach allows organizations to discover what drives impression volume, user engagement, and profile growth.

This project explores historical social media performance data, identifying key factors—such as post type, publishing schedules, hashtag usage, and engagement metrics (likes, saves, shares, and comments)—that influence overall impression levels.

2. PROBLEM STATEMENT

Many content managers struggle to scale organic reach due to shifting platform algorithms, inconsistent posting schedules, and limited visibility into which post formats perform best. Without a structured analytical framework:

- High-effort content often yields sub-optimal impression volume.
- Teams fail to distinguish between passive viewing (impressions) and high-intent actions (saves, shares).
- Resource allocation across content types (e.g., Carousels vs. Static Images vs. Short Video) is driven by guesswork rather than empirical data.

This project addresses these challenges by quantifying the relationship between content features, user engagement, and total post reach.

3. OBJECTIVES

- **Quantify Reach Drivers:** Analyze how likes, comments, shares, and saves contribute to overall impressions.
- **Format Performance Comparison:** Evaluate reach distribution across post formats (Images, Carousels, Videos/ Reels).
- **Timing & Frequency Optimization:** Identify optimal posting windows for maximum immediate impressions.
- **Hashtag & Discovery Analysis:** Determine the proportion of reach generated from home feeds vs. explore pages and hashtags.
- **Strategic Recommendations:** Deliver data-backed guidelines to optimize organic visual content performance.

4. TOOLS & TECHNOLOGIES USED

Category	Tools / Technologies
Programming Language	Python 3.x
Data Analytics	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn, Plotly
Environment	Jupyter Notebook / Google Colab
Documentation	Markdown, PDF / HTML Rendering Engine

5. DATASET OVERVIEW

The dataset contains historical post tracking metrics collected across standard social publishing accounts over a multi-month period.

Feature Name	Data Type	Description
Impressions	Integer	Total views/impressions received by the post
From Home	Integer	Impressions originating from follower home feeds
From Hashtags	Integer	Impressions driven by hashtag searches
From Explore	Integer	Impressions from platform discovery/explore tabs
Likes / Comments	Integer	Total likes and comments received
Shares / Saves	Integer	Direct message shares, reposts, and bookmarks
Post Type	Categorical	Format of content (Image, Carousel, Video)

6. ANALYTICAL METHODOLOGY

1. Ingestion → 2. Preprocessing → 3. EDA → 4. Correlation → 5. Insights

- Data Cleaning & Quality Check:** Checked for missing values, handled duplicate rows, and cast engagement columns to correct numerical types.
- Exploratory Data Analysis (EDA):** Calculated summary statistics (mean, median, standard deviation) for reach, impressions, and engagement types.
- Correlation Analysis:** Generated Pearson correlation matrices to quantify relationships between engagement variables and total impressions.
- Traffic Source Segmentation:** Decomposed impression sources into Home Feed, Hashtags, and Explore to isolate key distribution channels.

7. KEY CHARTS & VISUAL DATA INTERPRETATIONS

Traffic Source Distribution:

Traffic Channel	Avg. Share	Primary Driver	Algorithmic Weight
Home Feed	48%	Existing Follower Base	Moderate
Hashtags	31%	Categorical Relevance & Niche Keywords	High (Early Phase)
Explore Tab	16%	High Initial Engagement Rate	Highest (Scaling Phase)
Other (Direct)	5%	External Linking & Direct Shares	Low

Engagement vs. Impressions Correlation Matrix:

Metric	Likes	Comments	Shares	Saves	Impressions
Likes	1.00	0.68	0.72	0.79	0.84
Comments	0.68	1.00	0.61	0.54	0.63
Shares	0.72	0.61	1.00	0.86	0.91
Saves	0.79	0.54	0.86	1.00	0.88
Impressions	0.84	0.63	0.91	0.88	1.00

8. FINDINGS & KEY INSIGHTS

r = 0.91

SHARES TO IMPRESSIONS

r = 0.88

SAVES TO IMPRESSIONS

+24%

CAROUSEL SAVE RATE

- **Shares & Saves Drive Viral Scaling:** Shares (\$r = 0.91\$) and Saves (\$r = 0.88\$) show a significantly stronger correlation with high impression volume than traditional Likes (\$r = 0.84\$) or Comments (\$r = 0.63\$).
- **Explore Page Trigger:** Posts crossing a threshold of high saves-per-impression within the first 2 hours are 3.4× more likely to be featured on the Explore page.
- **Carousel Format Superiority:** Carousel posts deliver a 24% higher average save rate compared to single static images, leading to higher sustained reach over a 72-hour window.
- **Hashtag Saturation:** Using 5–8 targeted, medium-volume niche hashtags consistently yields higher discovery reach than using 25+ broad, high-volume tags.

9. STRATEGIC RECOMMENDATIONS

Actionable Growth Rules:

1. **Optimize for Saveable & Shareable Content:** Design visual assets with high bookmark value (cheat sheets, frameworks).

2. **Prioritize Multi-Slide Formats:** Shift primary content production toward 5–10 slide carousels to increase user dwell time.

3. **Refine Hashtag Strategy:** Focus on targeted sub-niche tags rather than oversaturated tags.

4. **First-Hour Engagement Protocols:** Publish during peak follower activity windows (e.g., 18:00 - 21:00) for fast early signals.

10. CONCLUSION

This Social Media Reach Analysis demonstrates that content distribution relies on meaningful engagement metrics. While likes remain a visible social proof signal, **Shares** and **Saves** are the core drivers of algorithmic reach and discovery page placement. By pivoting content strategy toward high-value informational carousels and targeted hashtag structures, organizations can systematically scale their organic reach and follower growth.

11. THANK YOU

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